College Physics 2

ELECTROMAGNETIC INDUCTION

MPS Department | FEU Institute of Technology

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Magnetic Flux & Faraday's Law Induction

MPS Department | FEU Institute of Technology

OBJECTIVES

- Discuss how a changing magnetic flux generates an electric field
- Describe the quantities used in electromagnetism
- Demonstrate how generator converts mechanical energy to electrical energy
- Explain how Faraday's law relates the induced emf in a loop to the change in magnetic flux through the loop.
- Calculate the emf induced in a conductor moving through a magnetic field

Electromagnetic Induction

Electromagnetic Induction is the process of using magnetic fields to produce voltage, and in a complete circuit, a current.

Michael Faraday first discovered it, using some of the works of Hans Christian Oersted. His work started at first using different combinations of wires and magnetic strengths and currents, but it wasn't until he tried moving the wires that he got any success.

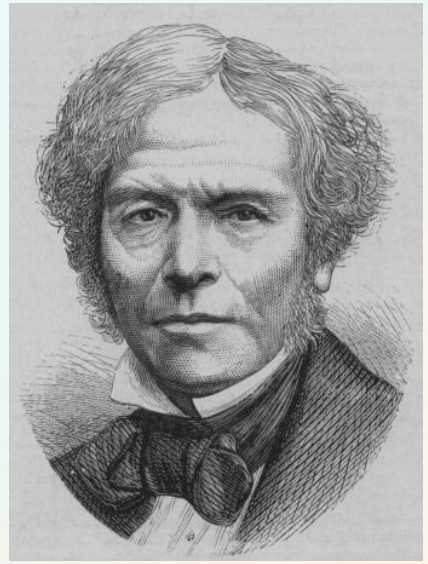


Figure 1. Michael Faraday. From. *https://tse1.mm.bing.net/th?id=OIP.39_6BXbofVeE_0AXgvYFwgAAAA&pid=Api&P=0&w=300&h=300*

It turns out that electromagnetic induction is created by just that - the moving of a conductive substance through a magnetic field.

ELECTROMAGNETIC INDUCTION

American Joseph Henry and Englishman Michael Faraday independently found that it is possible that a magnetic field can produce an electric current.

Henry made the discovery first but Faraday published his results earlier and investigated the subject in more detail.

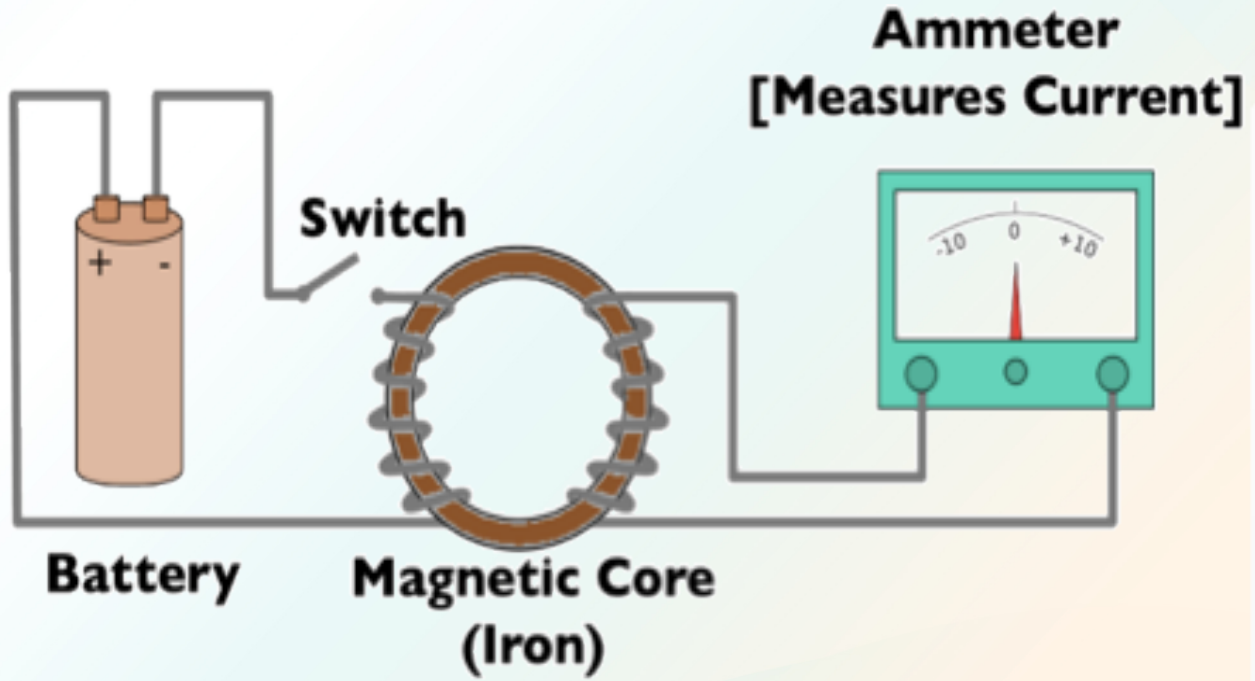


Figure 2. Induced emf. From. *http://hep0.okstate.edu/flera/phys1214/Ch21_Giancoli7e.pdf*

ELECTROMAGNETIC INDUCTION

A coil of wire X was connected to a battery. The current that flows through X produced a magnetic field that was intensified by the iron core around which the wire was wrapped.

The 2nd circuit Y contained a galvanometer to detect any current but contained no battery.

A constant current in X produced a constant magnetic field which produced NO CURRENT in Y. Only when the current in X was starting or stopping was a current produced in Y.

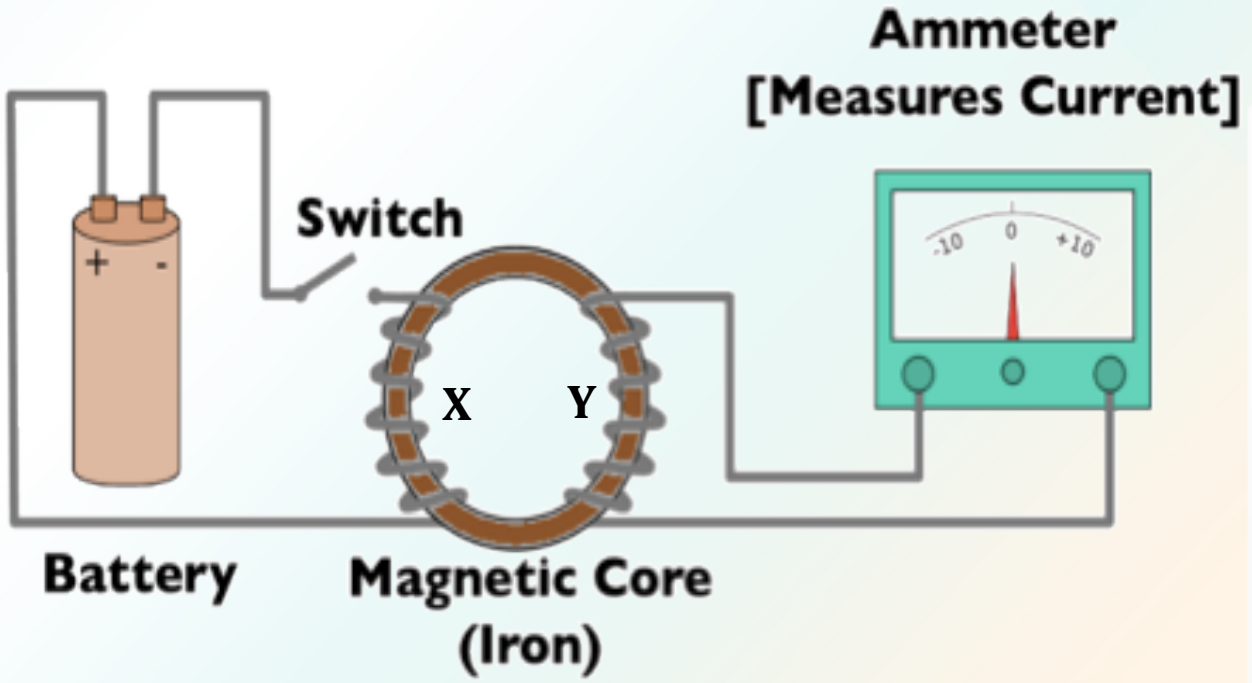


Figure 2. Induced emf. From. *http://hep0.okstate.edu/flera/phys1214/Ch21_Giancoli7e.pdf*

INDUCED EMF

Faraday concluded that although a constant magnetic field produces no current in a conductor, a changing magnetic field can produce an electric current. Such current is called an induced current. When the magnetic field through coil Y changes, a current occurs in Y as if there were a source of emf in circuit Y.

A changing magnetic field induces an emf.

Motion or change is required to induce emf. It does not matter whether the magnet or coil moves.

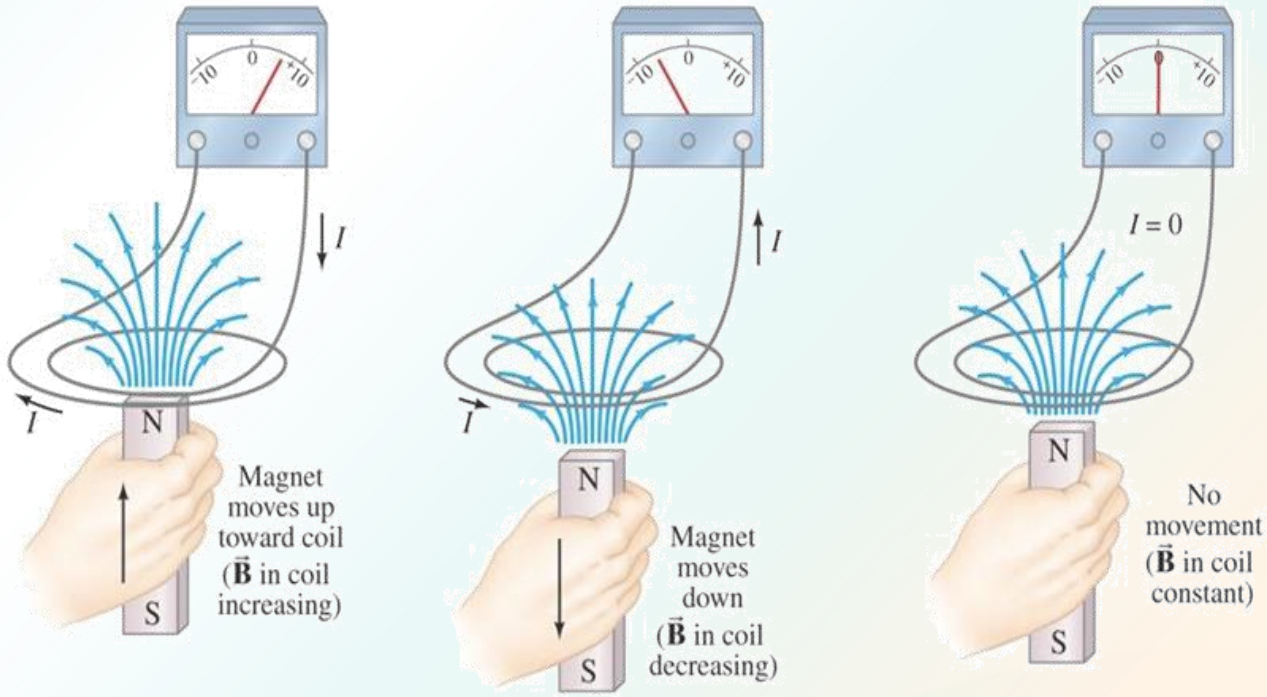


Figure 3. Changing magnetic field. From. *http://hep0.okstate.edu/flera/phys1214/Ch21_Giancoli7e.pdf*

MAGNETIC FLUX

The induced emf in a wire loop is proportional to the rate of change of the magne tic flux, Φ_B, through the loop.

$$\Phi_B = B_\perp A = BA\cos\theta$$

Where Φ_B = magnetic flux, Weber (Wb) (1 Wb= 1 Tm^2)

$B_\perp$ = component of the magnetic field B perpendicular to the face of the loop, T

B = uniform magnetic field, Tesla (T)

A = area of the loop, square meter (m^2)

θ = angle between B and a line perpendicular to the face of the loop

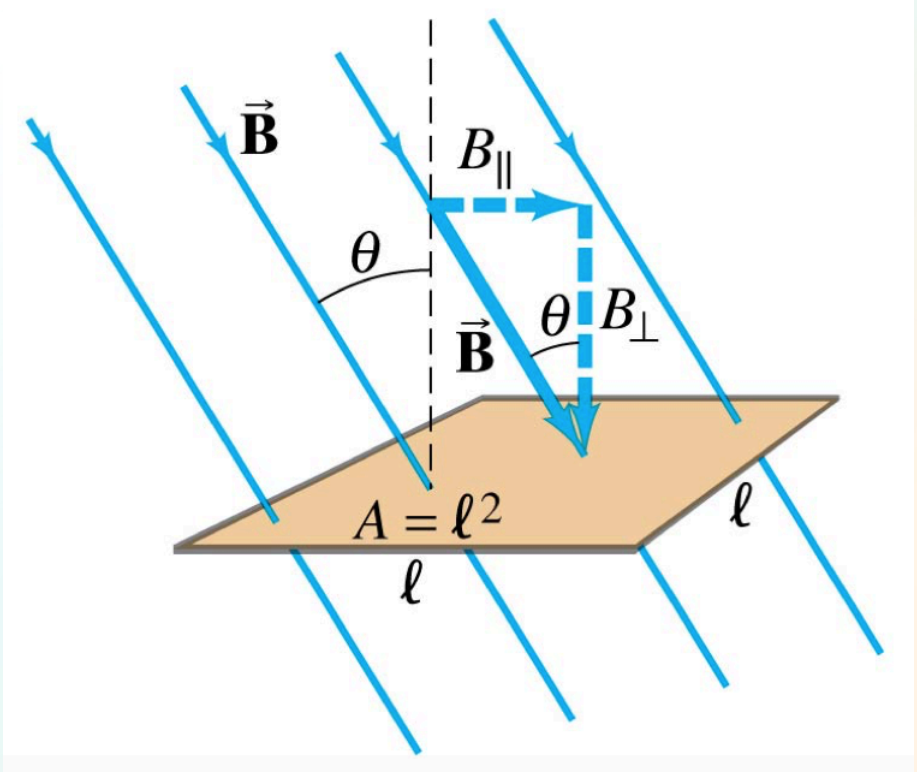


Figure 4. Changing magnetic field. From. *http://hep0.okstate.edu/flera/phys1214/Ch21_Giancoli7e.pdf*

MAGNETIC FLUX

Magnetic flux is proportional to the total number of lines passing through the loop

Ways to change the magnetic flux:
-Changing the magnitude of the field within a conducting loop
-Changing the area of the loop that lies within the magnetic field
-Changing the relative orientation of the field and the loop

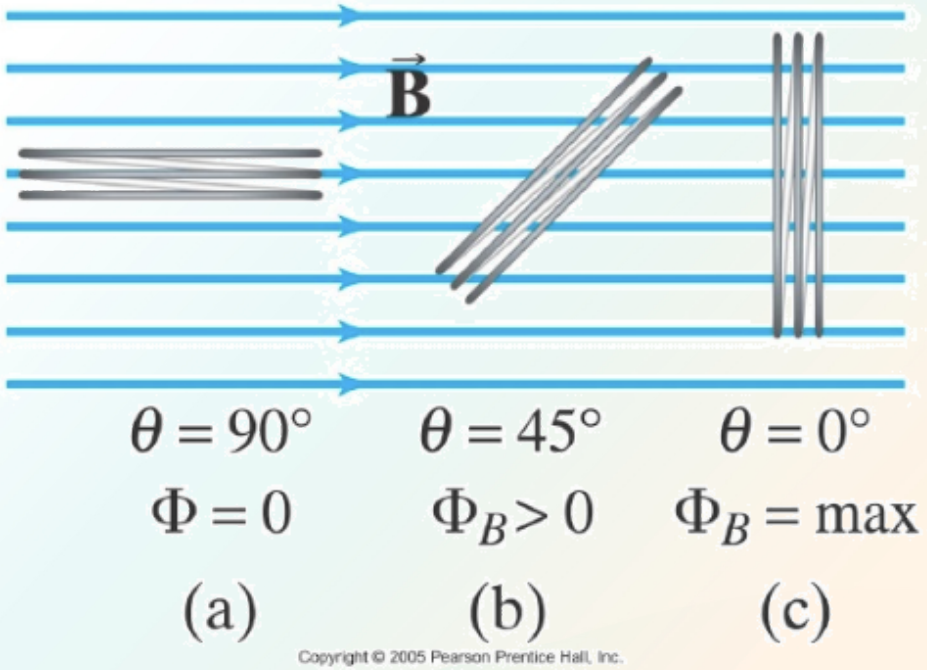
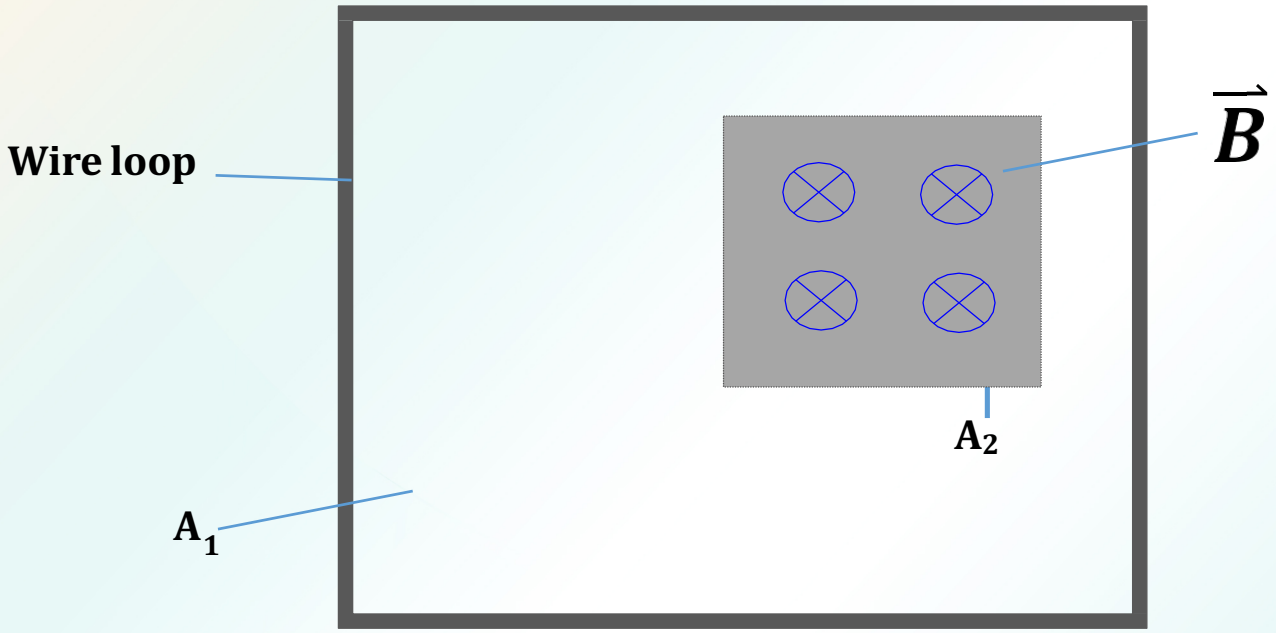


Figure 5. Changing magnetic field. From. *http://hep0.okstate.edu/flera/phys1214/Ch21_Giancoli7e.pdf*

Sample Problem

1. A square loop of wire encloses area A_1. A uniform magnetic field B perpendicular to the loop extends over the are A_2. What is the magnetic flux through the loop A_1?



$$\Phi_B = BA\cos\theta$$

$$\theta = 0^0$$

$$\Phi_B = BA_2 + B(A_1 - A_2)$$ (0)

$$\boldsymbol{\Phi_B = BA_2}$$

There is no magnetic field (B=0) in the area (A_1-A_2), so flux is zero in that area. Magnetic flux through loop A_1 is not BA_1 because B is NOT UNIFORM over A_1.

Figure 6. Square loop of wire. From. *https://image1.slideserve.com/3566120/magnetic-flux3-l.jpg*

Sample Problem

2. A square loop of wire 10.0 cm on a side is in a 1.25 T magnetic field B. What are the maximum and minimum values of flux that can pass through the loop?

$$\Phi_B = BAcos\theta$$

$$\Phi_B \; is \; \text{maximum} \; when \; \theta = 0^0$$

$$\Phi_{B,max} = (1.25T)(0.10m)(0.10m)cos0^0$$

$$\boldsymbol{\Phi_{B,max} = 0.0125\, Wb}$$

$$\Phi_B \; is \; \text{minimum} \; when \; \theta = 90^0$$

$$\Phi_{B,min} = BAcos0^0$$

$$\boldsymbol{\Phi_{B,min} = 0\, Wb}$$

FARADAY'S LAW

One of the basic laws of electromagnetism

$$\xi = -\frac{\Phi_B}{\Delta t}$$

If the circuit contains N closely wrapped loops, the emfs induced in each loop add together.

$$\xi = -N\frac{\Phi_B}{\Delta t}$$

- The minus sign gives the direction of the induced emf.

> **Faraday's law of induction states that the emf induced in a circuit is equal to the rate of change of magnetic flux through the circuit.**

FARADAY'S LAW

Faraday learned that if you change any part of the flux over time you could induce a current in a conductor and thus create a source of EMF (voltage, potential difference).

Since we are dealing with time here we are talking about the RATE of CHANGE of FLUX, which is called

Faraday's Law.

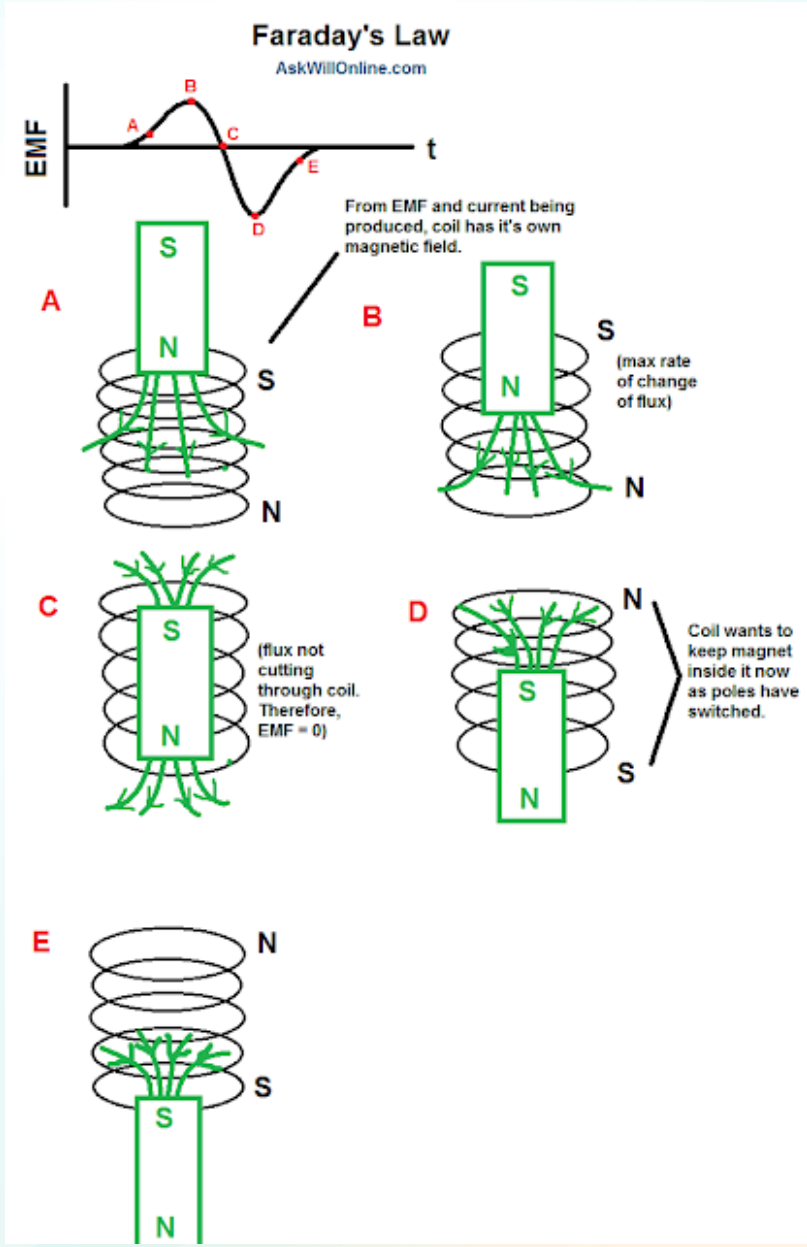


Figure 7. Rate of change of flux. From. *http://1.bp.blogspot.com/-73rqafR5TMs/UbRNfY9uChI/AAAAAAAAFsg/BmvOWfJ58gI/s640/Faraday*

APPLICATION

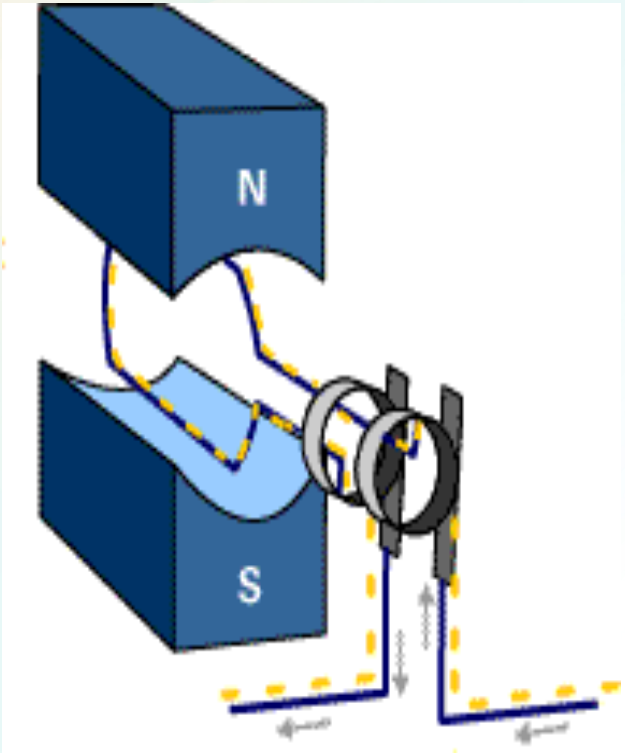


Figure 8. AC generator. From. *https://www.splung.com/fields/images/induction/acgen.gif*

AC Generators use Faraday's law to produce rotation and thus convert electrical and magnetic energy into rotational kinetic energy.

This idea can be used to run all kinds of motors.

Since the current in the coil is AC, it is turning on and off thus creating a CHANGING magnetic field of its own. Its own magnetic field interferes with the shown magnetic field to produce rotation.

APPLICATION

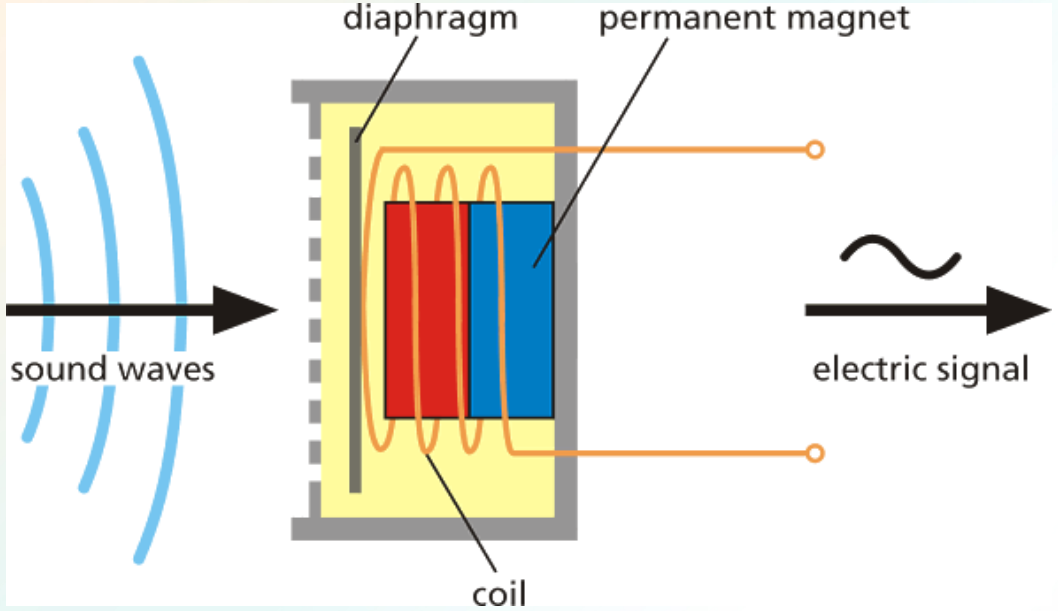


Figure 9. Microphone diaphragm. From. *http://www.pjaviation.co.za/wp-content/uploads/2012/05/image001.gif*

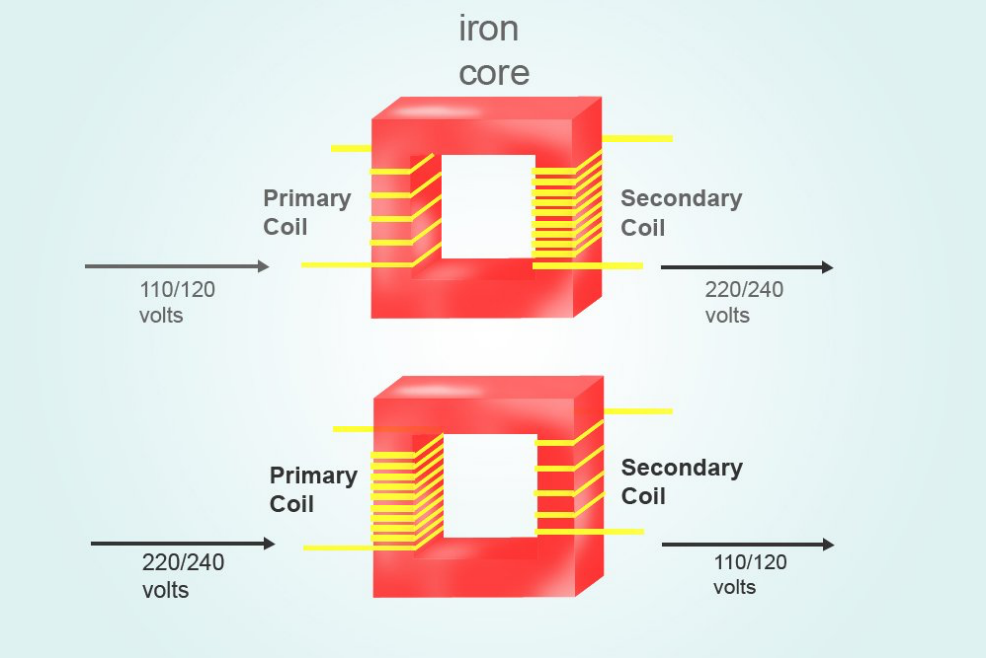
A **microphone** works when sound waves enter the filter of a microphone. Inside the filter, a diaphragm is vibrated by the sound waves which in turn moves a coil of wire wrapped around a magnet.

The movement of the wire in the magnetic field induces a current in the wire.

Thus sound waves can be turned into electronic signals and then amplified through a speaker.

APPLICATION

Probably one of the greatest inventions of all time is the transformer. AC Current from the primary coil moves quickly BACK and FORTH (thus the idea of changing!) across the secondary coil. The moving magnetic field caused by the changing field (flux) induces a current in the secondary coil.



If the secondary coil has MORE turns than the primary you can step up the voltage and runs devices that would normally need MORE voltage than what you have coming in. We call this a STEP UP transformer.

We can use this idea in reverse as well to create a STEP DOWN transformer.

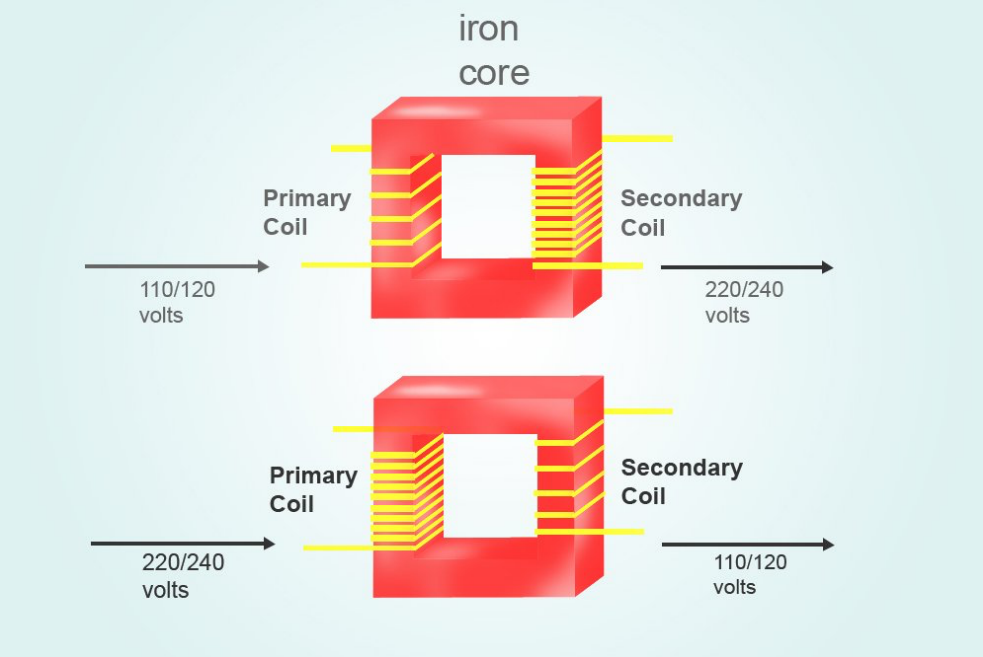
Figure 10. Transformer. From. *https://www.scienceabc.com/wp-content/uploads/ext-www.scienceabc.com/wp-content/uploads/2018/06/step-down-transformer.jpg-.jpg*

TRANSFORMER

From Faraday's law, the emf in the secondary and primary coil is

$$\varepsilon_s = N_s \frac{\Delta \emptyset_B}{\Delta t}$$

$$\varepsilon_p = N_p \frac{\Delta \emptyset_B}{\Delta t}$$



where

Ns is the number or turns in the secondary coil

Np is the number or turns in the secondary coil

Δφ/Δt is the rate at which the magnetic flux changes

$$\frac{\varepsilon_s}{\varepsilon_p} = \frac{N_s}{N_p}$$

Figure 10. Transformer. From. *https://www.scienceabc.com/wp-content/uploads/ext-www.scienceabc.com/wp-content/uploads/2018/06/step-down-transformer.jpg-.jpg*

Sample Problems

1. A coil with 200 turns of wire is wrapped on an 18.0 cm square frame. Each turn has the same area, equal to that of the frame, and the total resistance of the coil is 2.0Ω . A uniform magnetic field is applied perpendicularly to the plane of the coil. If the field changes uniformly from 0 to 0.500 T in 0.80 s, find the magnitude of the induced emf in the coil while the field has changed as well as the magnitude of the induced current.

Answer: emf = 4.1 V

Sample Problems

2. A transformer for home use of a portable radio reduces 120-V ac to 9.0-V dc. A secondary coil contains 30 turns and the radio draws 400mA. Calculate a) the number of turns in the primary coil, b) the current in the primary and c) the power transformed

Answer:

a) 400 turns
b) 0.03 A
c) 3.6 W

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

Giancoli, Douglas C. (2007). Physics 6th Edition.Jurong, Singapore. Pearson Education South Asia Pte Ltd.

https://tse1.mm.bing.net/th?id=OIP.39_6BXbofVeE_0AXgvYFwgAAAA&pid=Api&P=0&w=300&h=300

REFERENCES

https://tse1.mm.bing.net/th?id=OIP.39_6BXbofVeE_0AXgvYFwgAAAA&pid=Api&P=0&w=300&h=300

https://image1.slideserve.com/3566120/magnetic-flux3-l.jpg

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https://www.scienceabc.com/wp-content/uploads/ext-www.scienceabc.com/wp-content/uploads/2018/06/step-down-transformer.jpg-.jpg

College Physics 2

Lenz's Law, Motional EMF, Inductance & RL Circuits

MPS Department | FEU Institute of Technology

OBJECTIVES

- Determine the direction of an induced emf.
- Calculate the emf induced in a conductor moving through a magnetic field.
- Discuss how a changing magnetic flux generates an electric field that is very different from that produced by an arrangement of charges.
- Analyze circuits that include both a resistor and an inductor.

LENZ'S LAW

Lenz's law is used to determine the direction of the (conventional) electric current induced in a loop due to a change in magnetic flux inside the loop.

To produce an induced current, you need:

a. A closed conducting loop

b. An external magnetic flux through the loop that is changing in time

> Lenz's law states that a current produced by an induced emf moves in a direction so that its magnetic field opposes the original change in the flux.

LENZ'S LAW

How to determine the direction of the induced current?

1. **Determine whether the magnetic flux ($\Phi = BA\cos\theta$) inside the loop is decreasing, increasing, or unchanged.**

2. **The magnetic field due to the induced current ($B_{induced}$):**
   a. **points in the SAME direction as the external field if the flux is DECREASING**
   b. **points in the OPPOSITE direction as the external field if the flux is INCREASING**
   c. **ZERO if the flux is not changing.**

3. Once you know the direction of the induced magnetic field, use the right hand rule (thumb & curl of fingers for B and I) to find the direction of the induced current.

4. Always keep in mind that there are 2 magnetic fields.
   1. **an external field whose flux must be changing if it is to induce an electric current, and**
   2. **a magnetic field produced by the induced current.**

LENZ'S LAW

Magnetic field of induced current opposes change in external flux, NOT necessarily opposing the external field.

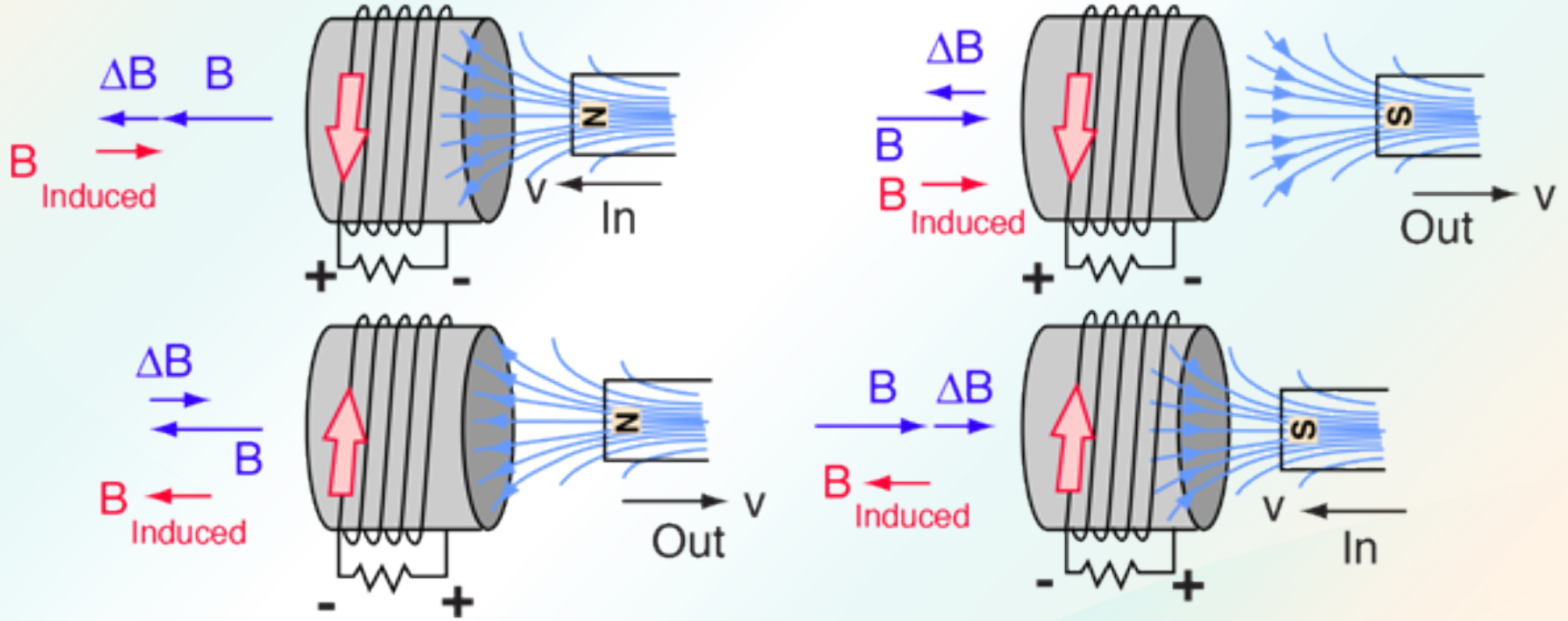
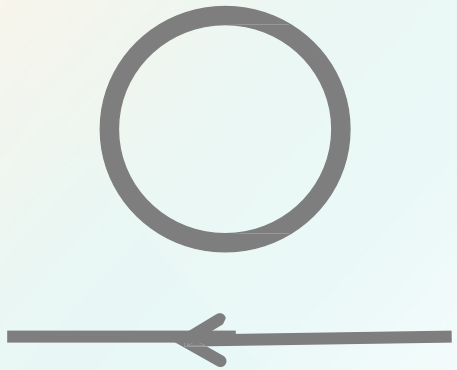


Figure 11. Lenz's Law. From. *https://tse1.mm.bing.net/th?id=OIP.xC6-glhcGg2ebKB7vRH5BAHaFj&pid=Api&P=0&w=259&h=195*

Sample Problems

1. What is the direction of the induced current in the circular loop due to the current shown in the figure.



I increasing

Figure 12. Sample 1 Lenz's Law. From. *http://d2vlcm61l7u1fs.cloudfront.net/media%2F502%2F502b46b0-933e-4374-b0e2-c2ba6eef9411%2FphpquDpjb.png*

Using RHR: **thumb** – current (**I**) direction (to the left)
curl of fingers – direction of magnetic field (**B**) in the loop is INWARD

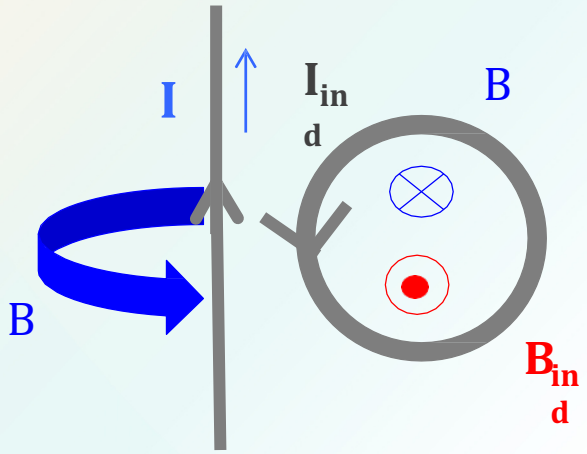
Since the current (I) is increasing, the magnetic field (B) is also increasing. Thus, the direction the induced magnetic field (B_{ind}) is OUTWARD (towards you). (Rule: The magnetic field due to the induced current ($B_{induced}$) points in the OPPOSITE direction as the external field if the flux is INCREASING)

Using RHR again: **thumb** – induced magnetic field (B_{ind}) in OUTWARD direction

COUNTERCLOCKWISE **curl of fingers** –induced current (I_{ind}) direction which is

Sample Problems

2. What is the direction of the induced current in the circular loop due to the current shown in the figure.



I decreasing

Figure 13. Sample 2 Lenz's Law. From. *http://d2vlcm61l7u1fs.cloudfront.net/media%2F502%2F502b46b0-933e-4374-b0e2-c2ba6eef9411%2FphpquDpjb.png*

Current (**I**) is to the LEFT
Magnetic field (**B**) in the loop is INWARD
Induced magnetic field (B_{ind}) is OUTWARD
Induced current (I_{ind}) is **COUNTERCLOCKWISE**

Sample Problems

3. What is the direction of the induced current in the circular loop due to the current shown in the figure.



I decreasing

Figure 14. Sample 3 Lenz's Law. From. *http://d2vlcm61l7u1fs.cloudfront.net/media%2F502%2F502b46b0-933e-4374-b0e2-c2ba6eef9411%2FphpquDpjb.png*

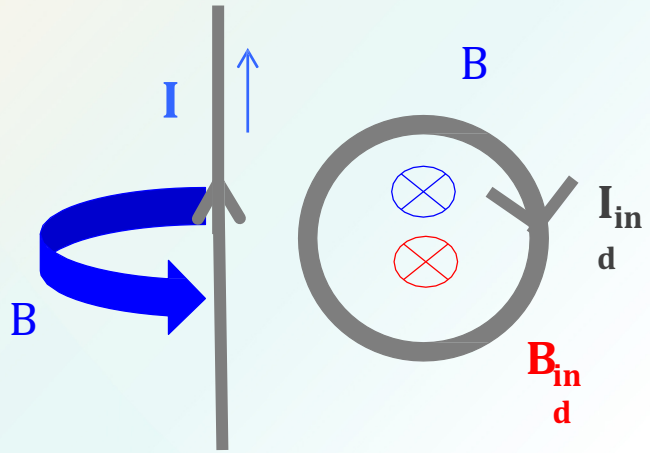
Using RHR: **thumb** – current (**I**) in UPWARD direction
curl of fingers – direction of magnetic field (**B**) in the loop is INWARD

Since the current (I) is decreasing, the magnetic field (B) is also decreasing. Thus, the direction the induced magnetic field (B_{ind}) is ALSO INWARD (into the page). (Rule: The magnetic field due to the induced current ($B_{induced}$) points in the SAME direction as the external field if the flux is DECREASING.)

Using RHR again: **thumb** – induced magnetic field (B_{ind}) in INWARD direction
curl of fingers –induced current (I_{ind}) direction which is **CLOCKWISE**

Sample Problems

4. What is the direction of the induced current in the circular loop due to the current shown in the figure.



I decreasing

Figure 15. Sample 4 Lenz's Law. From. *http://d2vlcm61l7u1fs.cloudfront.net/media%2F502%2F502b46b0-933e-4374-b0e2-c2ba6eef9411%2FphpquDpjb.png*

Current (**I**) is UPWARD
Magnetic field (**B**) in the loop is INWARD
Induced magnetic field (B_{ind}) is ALSO INWARD
induced current (I_{ind}) is **CLOCKWISE**

EMF INDUCED IN A MOVING CONDUCTOR

A uniform magnetic field B is perpendicular to the area bounded by the U-shaped conductor and the movable rod resting on it.

If rod is made to move at a speed of v, it travels a distance of Δx = vt in a time Δt. The area of the loop increases by an amount ΔA=lΔx = lvΔt in a time Δt. By Faraday's law, there is an induced emf whose magnitude is:

$$\xi = \frac{\Phi_B}{\Delta t} = \frac{B\Delta A}{\Delta t} = \frac{B\ell v\Delta t}{\Delta t} = B\ell v$$

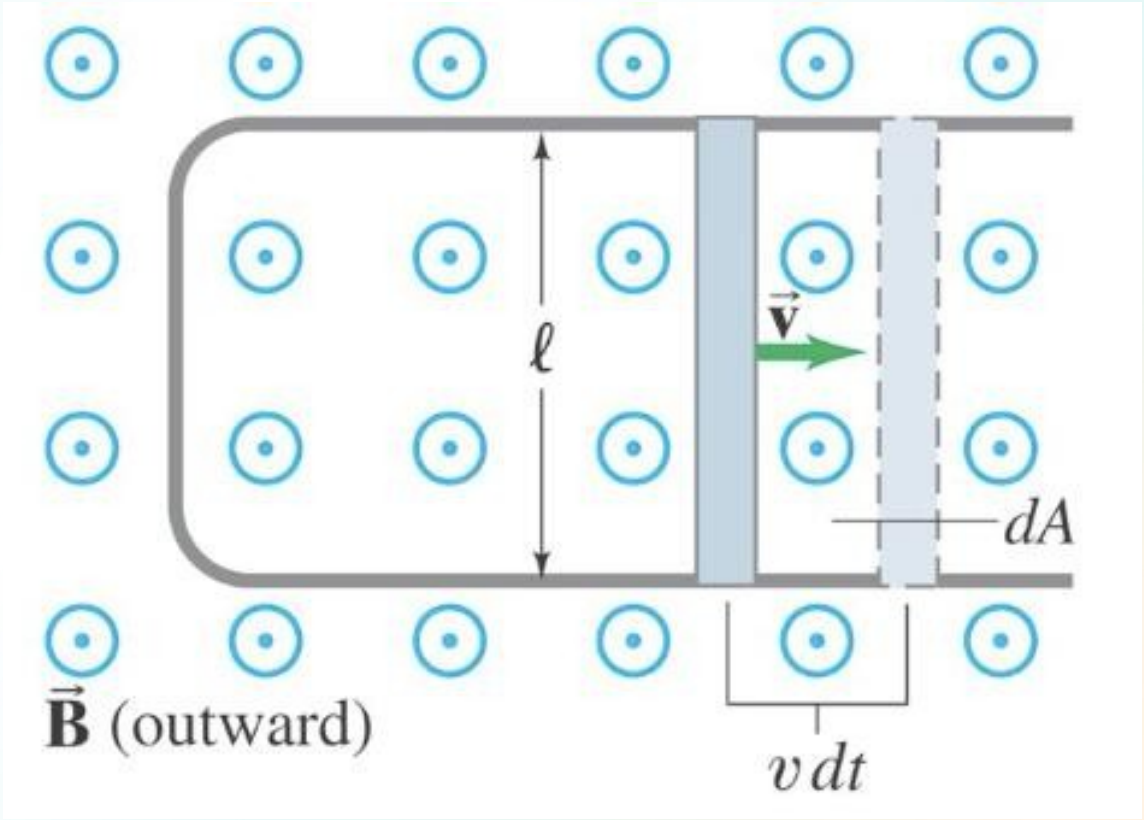


Figure 16. A conducting rod is moved to the right on a U-shaped conductor in a uniform magnetic field B that points out of the paper. From. *http://hep0.okstate.edu/flera/phys1214/Ch21_Giancoli7e.pdf*

EMF INDUCED IN A MOVING CONDUCTOR

$$\xi = \frac{\Phi_B}{\Delta t} = \frac{B\Delta A}{\Delta t} = \frac{B\ell v\Delta t}{\Delta t} = B\ell v$$

Equation is valid as long as B, l, v are mutually perpendicular.

An emf induced on a conductor by moving in a magnetic field is called motional emf.

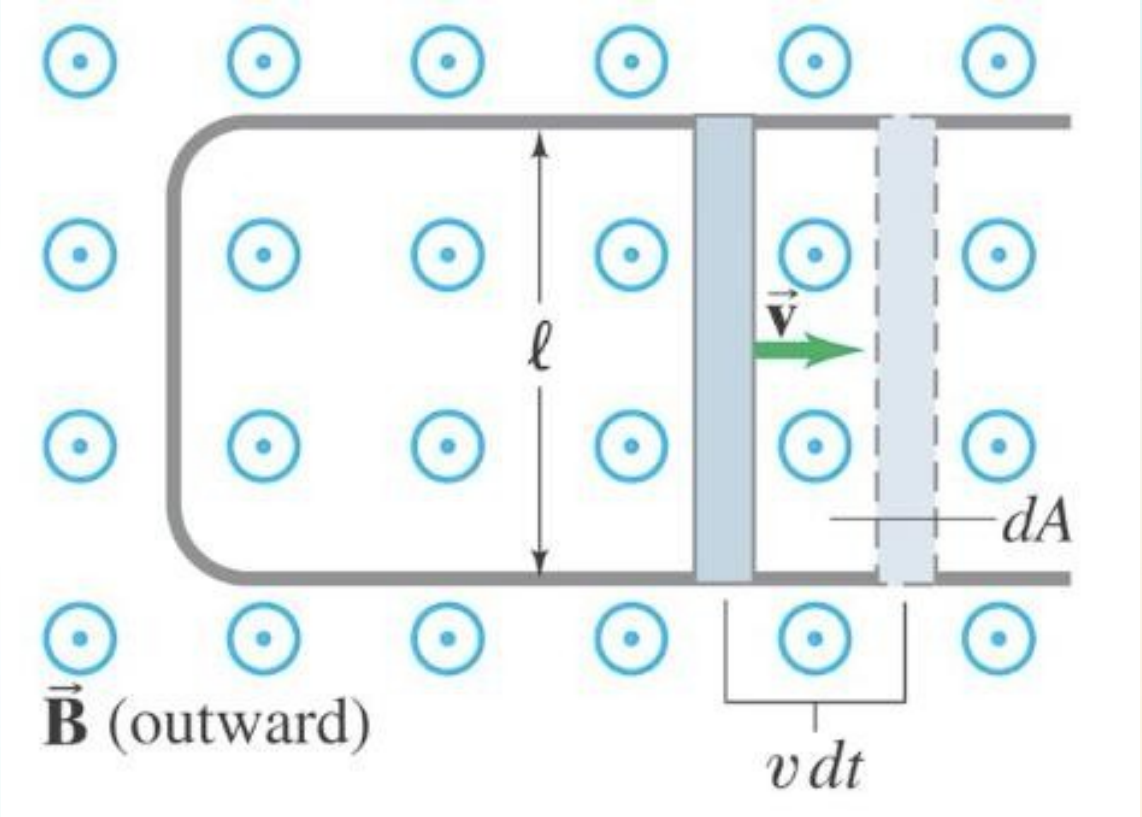


Figure 16. A conducting rod is moved to the right on a U-shaped conductor in a uniform magnetic field B that points out of the paper. From. *http://hep0.okstate.edu/flera/phys1214/Ch21_Giancoli7e.pdf*

EMF INDUCED IN A MOVING CONDUCTOR

The induced current is in a direction that tends to slow the moving bar. It takes an external force to keep it moving.

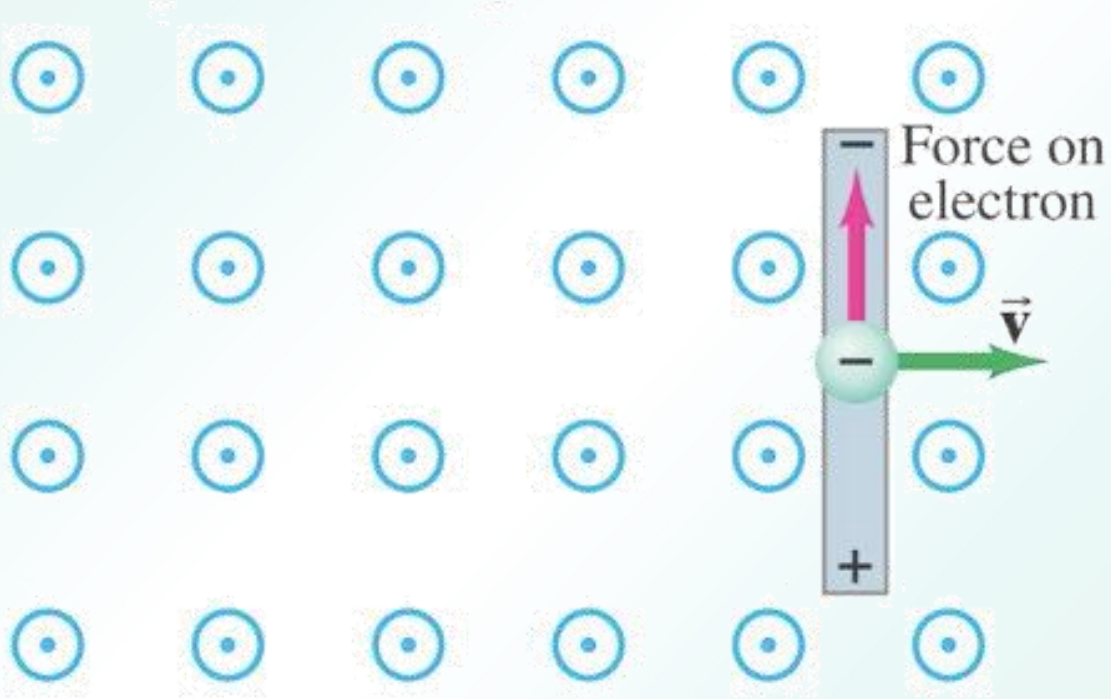


Figure 17. Induced current tends to slow the moving bar. From. *http://hep0.okstate.edu/flera/phys1214/Ch21_Giancoli7e.pdf*

CHANGING MAGNETIC FLUX PRODUCING ELECTRIC FIELD

A changing magnetic field induces an electric field.

This applies not only to wires and other conductors, but a general result that applies to any region in space: an electric field will be induced at any point in space where there is a changing magnetic field.

For the case of electrons in a moving conductor, a simple formula for E in terms of B. The electric field in the rod is:

$$E = \frac{F}{q} = \frac{qvB}{q} = vB$$

Sample Problems

5. An airplane travels 1000km/h in a region where the Earth's magnetic field is 5.0 x10^-5 T and is nearly vertical. What is the potential difference induced between the wing tips that are 70m apart? (Consider the wings to be a 70m long conductor moving through the Earth's magnetic field.)

Solution:

$$v = \left(\frac{1000km}{h}\right)\left(\frac{1h}{3600s}\right)\left(\frac{1000m}{km}\right)$$

$$v = 280m/s$$

$$\xi = B\ell v$$

$$\xi = \left(5x10^{-5}T\right)(70m)\left(\frac{280m}{s}\right)$$

$$\xi = 1.0\,V$$

EDDY CURRENTS

Eddy currents can dramatically slow a conductor moving into or out of a magnetic field.

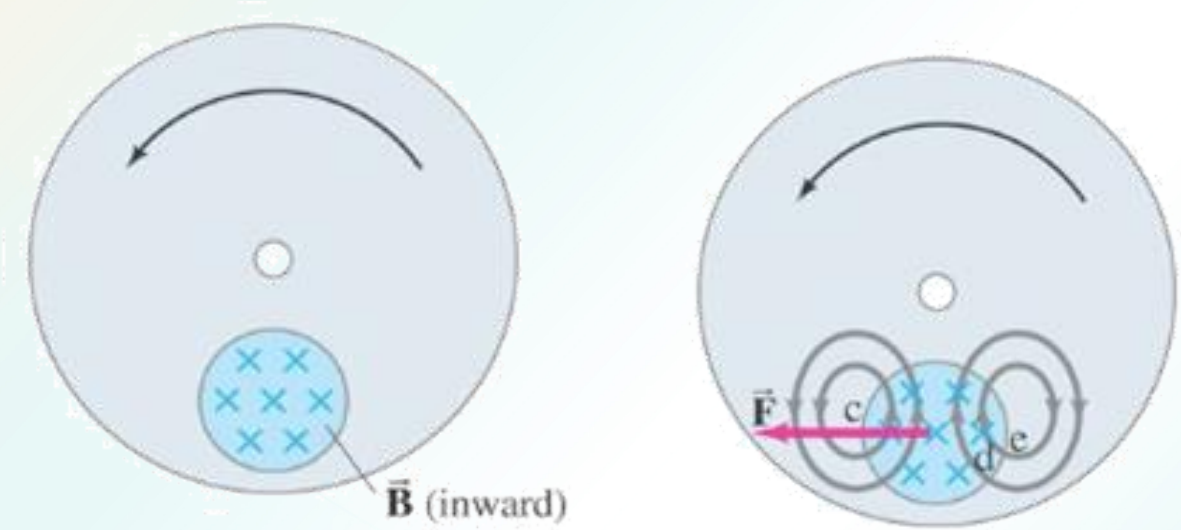


Figure 18. Production of Eddy currents in a rotating wheel. From. *http://image.slideserve.com/228996/eddy-current-l.jpg*

> **Eddy currents are induced currents flowing in bulk material as well through wires.**

INDUCTANCE

Mutual Inductance

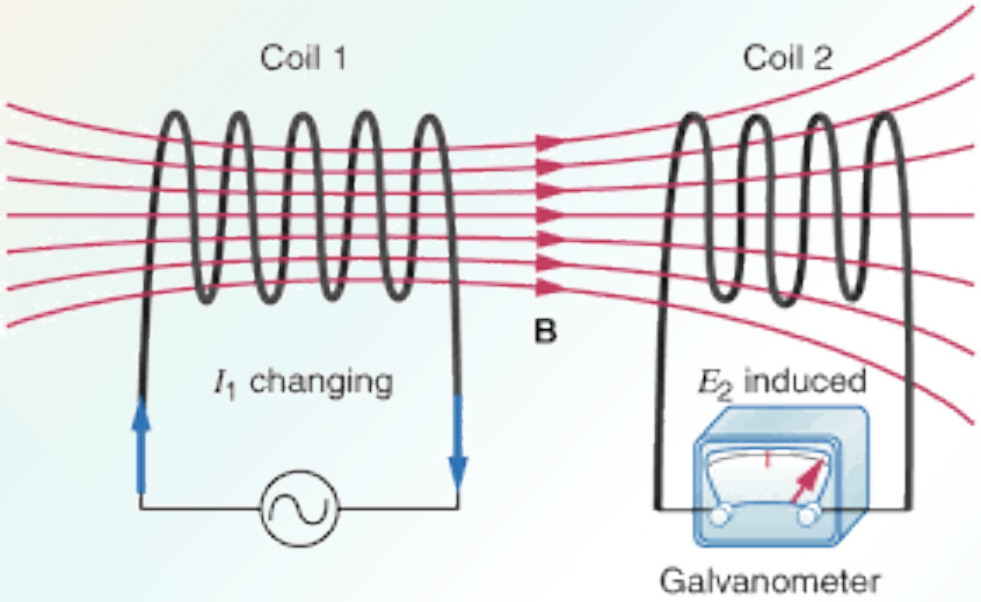


Figure 19. Mutual Inductance. From. *https://tse4.mm.bing.net/th?id=OIP.QtNMipESQn7FAEsvcWx0cwHaEs&pid=Api&P=0&w=241&h=154*

> **Mutual inductance is the phenomenon in which a change in current in one coil causes an induced current in another coil placed near to the first coil.**

$$\xi_1 = -\frac{M\Delta I_2}{\Delta t}$$

$$\xi_2 = -\frac{M\Delta I_1}{\Delta t}$$

INDUCTANCE

$$\xi_1 = -\frac{M\Delta I_2}{\Delta t}$$

$$\xi_2 = -\frac{M\Delta I_1}{\Delta t}$$

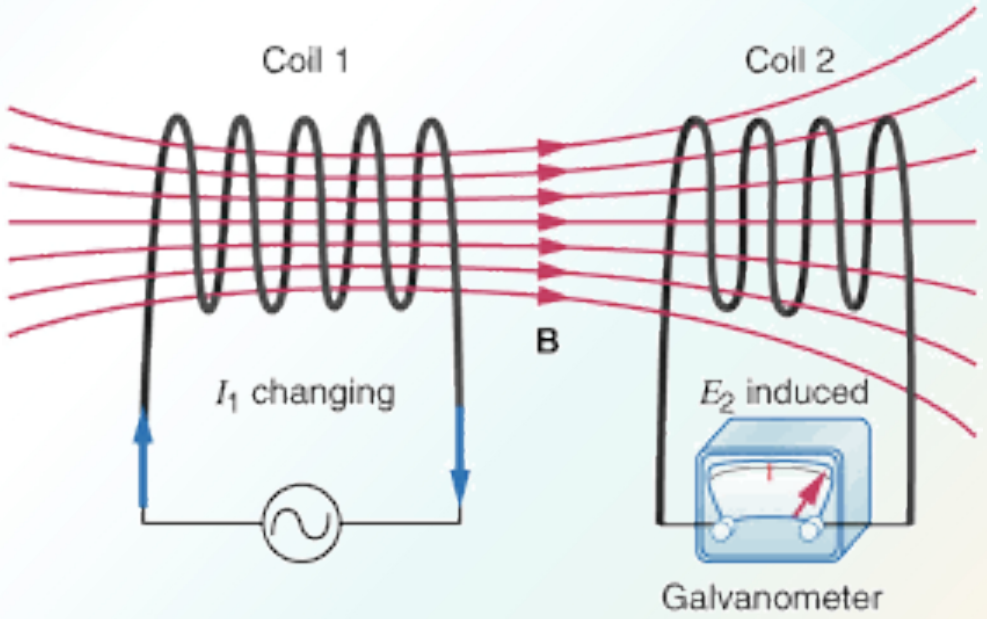


Figure 19. Mutual Inductance. From. *https://tse4.mm.bing.net/th?id=OIP.QtNMipESQn7FAEsvcWx0cwHaEs&pid=Api&P=0&w=241&h=154*

Where M = mutual inductance, henry (H)

1H = 1 Vs/A = 1 Ωs

I =current, A

t = times, s

ξ =emf

M does not depend on I1. It depends on the geometric factors such as size, shape, number of turns and relative positions of the 2 coils and also on whether iron (or other ferromagnetic material) is present.

Applications:

-transformers

-pacemakers

INDUCTANCE

If the current in the coil is changing, the changing flux through the coil induces an emf in the coil.

$$\xi = -\frac{L\Delta I}{\Delta t}$$

Where

L = self-inductance, henry (H)

1H = 1 Vs/A = 1 Ωs

I =current, A

t = times, s

ξ =emf

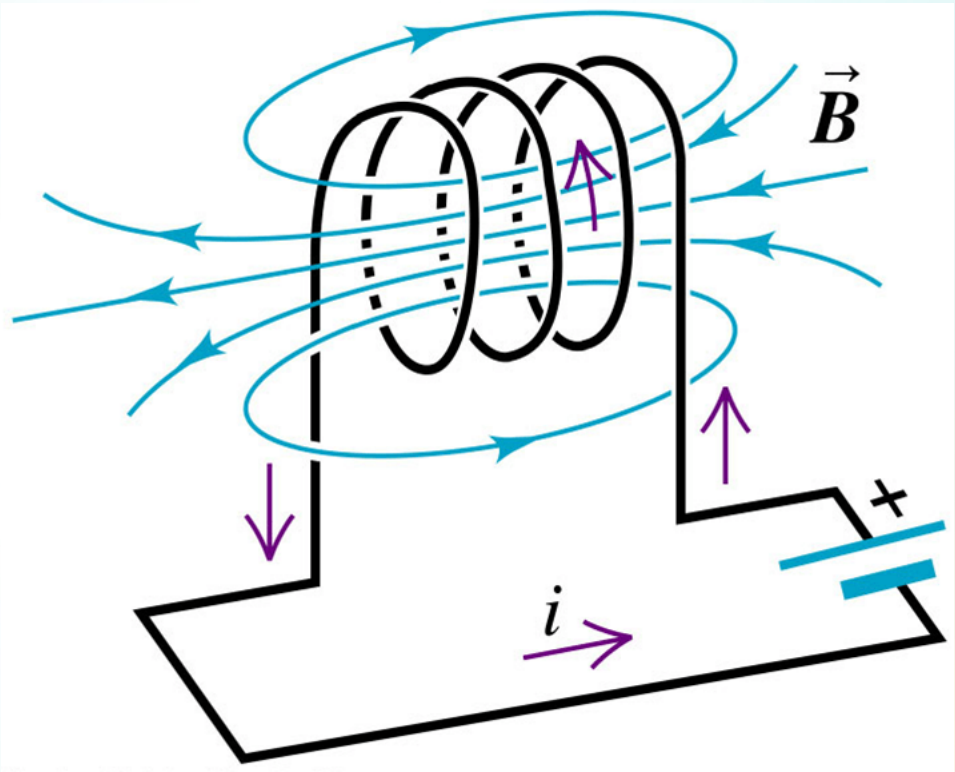


Figure 20. Inductance. From. *https://tse1.mm.bing.net/th?id=OIP.3a0mzli2W76Cq5UXIY33CgHaF9&pid=Api&P=0&w=222&h=179*

INDUCTANCE

SELF-INDUCTANCE

$$\xi = -\frac{L\Delta I}{\Delta t}$$

The magnitude of L depends on the size and the shape of the coil and on the presence of an iron core.

An ac circuit always contains some inductance, but quite small unless the circuit contains a coil of many loops or turns.



Figure 21. Several inductors. From. *https://tse3.mm.bing.net/th?id=OIP.AMreBf6ibGzO1lf6WQFeLgHaGk&pid=Api&P=0&w=184&h=164*



Figure 22. Inductor symbol.From. *https://tse1.mm.bing.net/th?id=OIP.-W_WoN1m90JCK3ZwNtvqnQAAAA&pid=Api&P=0&w=300&h=300*

A coil that has a significant self- inductance (L) is called an **inductor**.

INDUCTANCE

The direction of a self-induced emf is found rom Lenz's law. The cause of the emf is an increasing or decreasing in current.

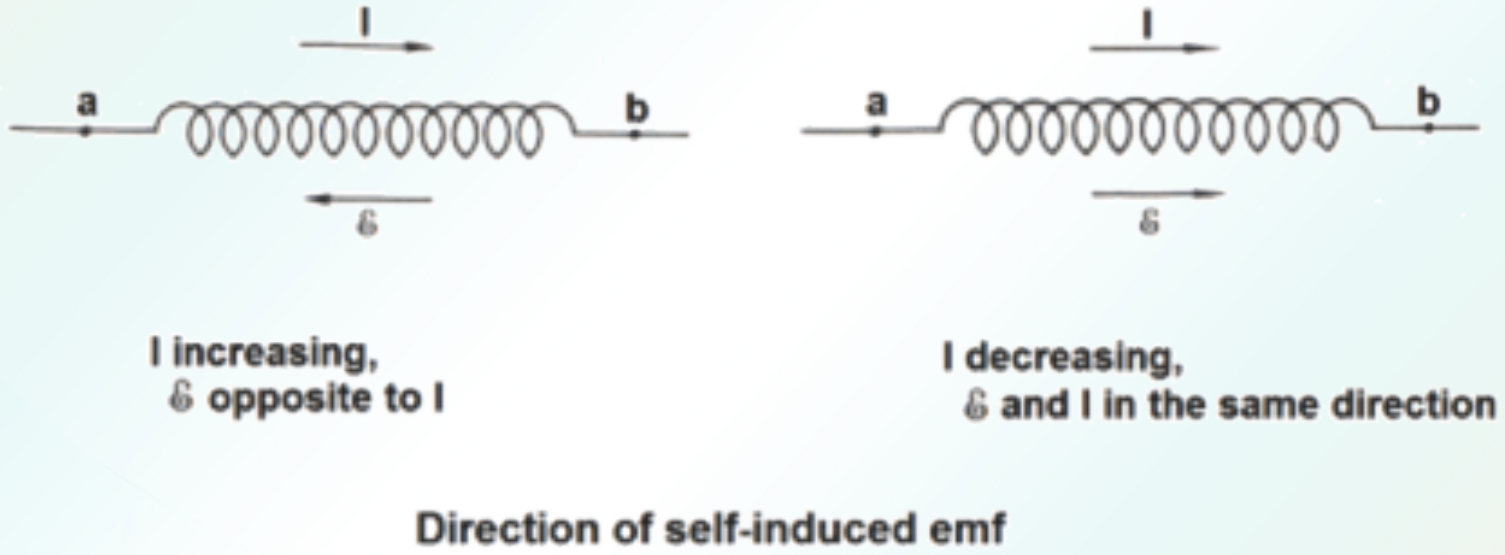


Figure 23. Direction of self-induced emf. From. *https://tse3.mm.bing.net/th?id=OIP.HXFyxSzuuOeXceDbfFbNEgHaDm&pid=Api&P=0&w=316&h=154*

RL CIRCUITS

Any inductor will have resistance. It is represented by drawing the inductance L and the resistance R separately. The resistance R could also include a separate resistor connected in series.

When the switch is turned on (closed), the current starts to flow. It is opposed by the induced emf in the inductor because of the changing current. However, as soon as the current starts to flow, there is a voltage drop across the resistance (V=IR).

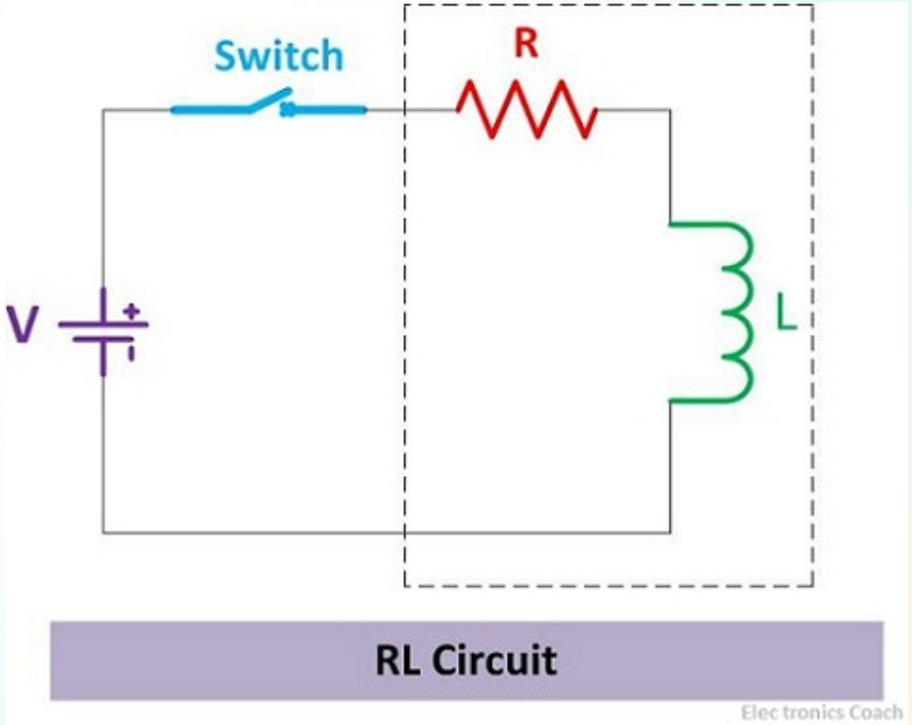


Figure 24. RL circuit. From. *https://electronicscoach.com/wp-content/uploads/2017/12/RL-circuit.jpg*

RL CIRCUITS

LR Circuit with emf

$$I = \left(\frac{V}{R}\right)\left(1 - e^{-t/\tau}\right)$$

LR Circuit without emf

$$I = I_{max}e^{-t/\tau}$$

$$\tau = \frac{L}{R}$$

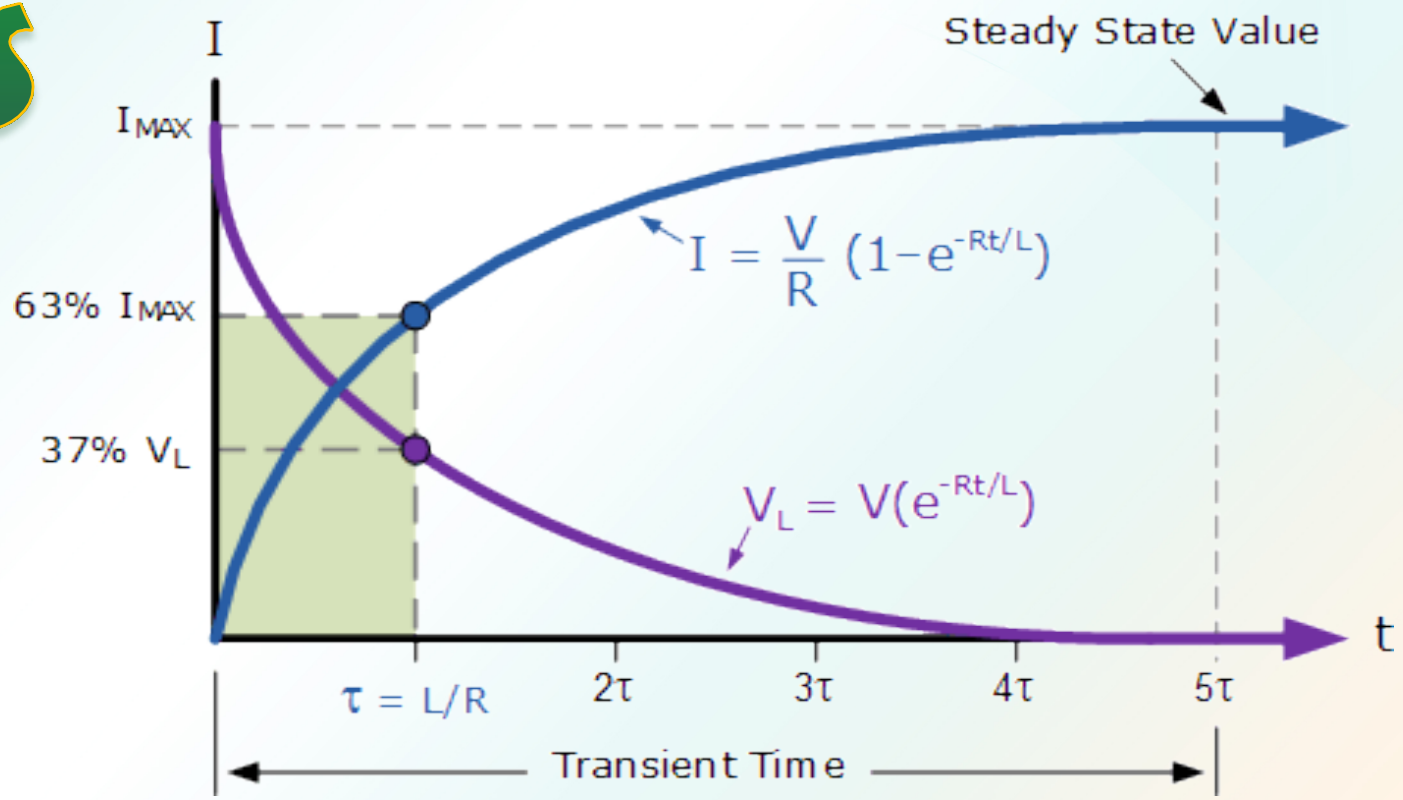


Figure 25. Transient time .From. *https://tse4.mm.bing.net/th?id=OIP.H5IrzNcRLvtdvJfQL2nCqQAAAA&pid=Api&P=0&w=256&h=148*

The time constant (τ) is the time for the current to decrease to 37% of the original value.
The time constant is inversely proportional to R.

Sample Problem

A solenoid has an inductance of 87.5mH and a resistance of 0.250Ω. Find (a) the time constant for this circuit, and (b) how long it would take for the current to go from zero to 63% of its final (maximum) value when connected to a battery of voltage, V.

Solution:

$$\tau = \frac{L}{R}$$

$$\tau = \frac{87.5x10^{-3}H}{0.250\Omega}$$

$$\tau = 0.35s$$

$$I = \left(\frac{V}{R}\right)\left(1 - e^{-t/\tau}\right)$$

$$I = 0.63\left(\frac{V}{R}\right)$$

$$0.63\left(\frac{V}{R}\right) = \left(\frac{V}{R}\right)\left(1 - e^{-t/\tau}\right)$$

$$0.63 = 1 - e^{-\frac{t}{0.35}}$$

$$e^{-\frac{t}{0.35}} = 1 - 0.63 = 0.37$$

$$\ln\left(e^{-\frac{t}{0.35}}\right) = \ln(0.37)$$

$$-\frac{t}{0.35} = \ln(0.37)$$

$$t = (-0.35)(ln0.37)$$

$$t = 0.35\,s$$

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

Giancoli, Douglas C. (2007). Physics 6th Edition.Jurong, Singapore. Pearson Education South Asia Pte Ltd.

https://tse1.mm.bing.net/th?id=OIP.xC6-glhcGg2ebKB7vRH5BAHaFj&pid=Api&P=0&w=259&h=195

REFERENCES

http://d2vlcm61l7u1fs.cloudfront.net/media%2F502%2F502b46b0-933e-4374-b0e2-c2ba6eef9411%2FphpquDpjb.png

https://tse4.mm.bing.net/th?id=OIP.QtNMipESQn7FAEsvcWx0cwHaEs&pid=Api&P=0&w=241&h=154

https://tse1.mm.bing.net/th?id=OIP.3a0mzli2W76Cq5UXIY33CgHaF9&pid=Api&P=0&w=222&h=179

https://tse3.mm.bing.net/th?id=OIP.AMreBf6ibGzO1lf6WQFeLgHaGk&pid=Api&P=0&w=184&h=164

REFERENCES

http://www.solitaryroad.com/c1048/ole12.gif

https://electronicscoach.com/wp-content/uploads/2017/12/RL-circuit.jpg

https://tse4.mm.bing.net/th?id=OIP.H5IrzNcRLvtdvJfQL2nCqQAAAA&pid=Api&P=0&w=256&h=148